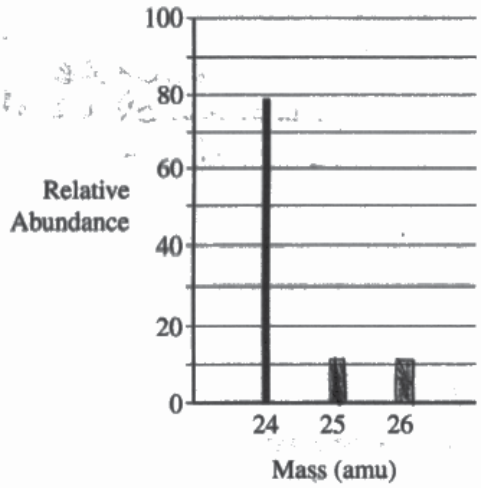
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Part A

Question 1



i. 1 − .79 = .21 ÷ 2 = .105 ≈ 10.5%

ii. Each isotope has a different number of neutrons resulting in each one having a different mass.

Question 1 is continued on the next page.

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Question 1

Part B

i. Na^+ has a weaker attraction because it has a smaller charge, since Mg has a charge of +2 it is looking to gain 2 negative charges compared to Na only needing 1.

ii. Na^+ is bigger than Mg^{+2} so the nucleus does not have as strong of a pull as Mg does.

Part C

$-\log(2.8\times10^{-4}) = pOH$

$pOH = 3.55$

$14 - 3.55 = 10.45$

$pH = 10.45$

Question 1 is continued on the next page.

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Question 1

Part D

1.85×10^{-3} $Mg(NO_3)_2 + 2NaOH \longrightarrow 2NaNO_3 + MgOH$

.155 6.475×10^{-5}

Part E

i. $K_{sp} = [Mg^{+2}][OH^-]^2$

ii. $Q = \frac{1.65\times10^{-4}}{6.475\times10^{-4}}$ $Q = .255$

iii. Yes a precipitate should form because $Q > K$ so the reverse reaction is favored and $Mg(OH)_2(s)$ will form.

Question 1 is continued on the next page.

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Question 1

Part F

The amount of undissolved $Mg(OH)_2$ should increase because the solution started as $Mg(NO_3)_2$ so adding HNO_3 would add a common ion and decrease the K_{sp}.

Continue to Question 2.